LITERATURE SURVEY · ML MINI PROJECT

IndicDoc AI

DocumentLayoutParsingforMultilingualIndianDocuments

1 2 3 4 5 6 7 8 **8 research papers**

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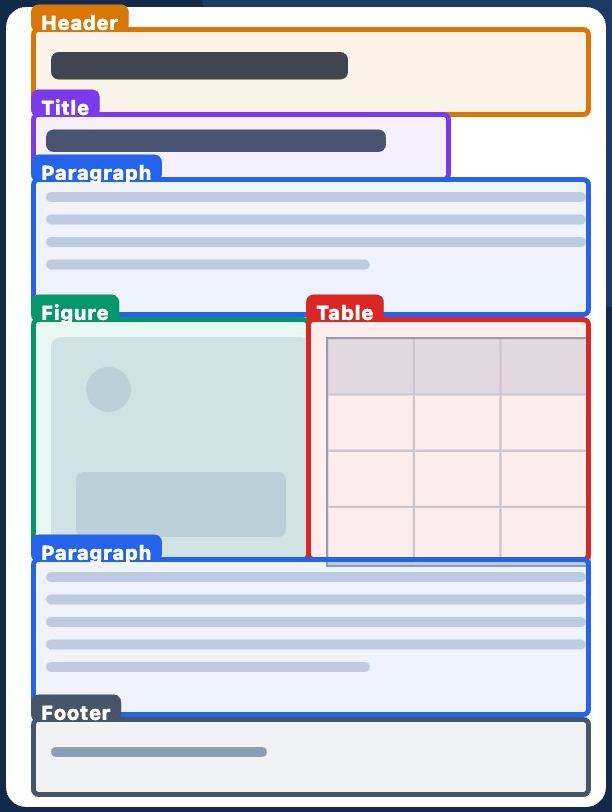
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Course: ML Mini Project

Guide: Dr. Joanne Gomes

Institute: St. Francis Institute of Technology



WHY THIS WORK

Motivation & SDG Mapping

WHY THIS PROBLEM MATTERS

Many languages, many scripts
22 scheduled languages; scripts have ligatures and diacritics.

Huge paper archives
Circulars, textbooks, newspapers, forms — often scanned.

Tools are built for English
Most layout & table datasets are English-only.

Key point: layout detection is step 1 — wrong regions mean wrong OCR, tables and answers.

MAPPED SDGS

SDG 4
Quality Education
Digital textbooks in Indian languages

SDG 9
Industry & Innovation
AI building blocks for Indic documents

SDG 10
Reduced Inequalities
Document AI beyond English

SDG 16
Strong Institutions
Searchable government records

WHAT WE SOLVE

Problem Statement

Problem: find and label every region on an Indian-language page — paragraphs, headings, tables, figures — so OCR, table extraction and QA can follow.



Input page

Labelled regions

SCOPE

- Indian-language pages (IndicDLP, via AIKosh)
- Region detection: boxes + labels
- YOLOv8s and Faster R-CNN
- OCR, table QA, grounding → future

RESEARCH OBJECTIVE

Compare one-stage (YOLOv8s) and two-stage (Faster R-CNN) detectors, using IndicDLP's YOLO results as reference.

SELECTED PAPERS

Paper Details

1 IndicDLP

IndicDLP: A Foundational Dataset for Multi-Lingual and Multi-Domain Document Layout Parsing

O. Nath, S. Kukkala, M. Khapra, R. K. Sarvadevabhatla

2025 arXiv preprint

IIT Madras & IIIT Hyderabad

2 PLATTER

PLATTER: A Page-Level Handwritten Text Recognition System for Indic Scripts

B. V. Kasuba, D. Kudale, V. Subramanian, P. Chaudhuri, G. Ramakrishnan

2025 arXiv preprint

IIT Bombay

3 UniTabBank

UniTabBank: A Large Scale Multi-Lingual, Multi-Layout, Multi-Type, Multi-Format Dataset for Table Detection

A. Mondal, S. Mundra, A. Dasgupta, C. V. Jawahar

2026 IEEE/CVF WACV

IIIT Hyderabad

4 Indic Table QA

Table Question Answering for Low-resourced Indic Languages

V. Pal, E. Kanoulas, A. Yates, M. de Rijke

2024 arXiv preprint

University of Amsterdam

5 LayoutXLM

LayoutXLM: Multimodal Pre-training for Multilingual Visually-rich Document Understanding

Y. Xu, T. Lv, L. Cui, G. Wang, Y. Lu, D. Florencio, C. Zhang, F. Wei

2021 arXiv preprint

Microsoft Research Asia

6 DRISHTIKON

DRISHTIKON: Visual Grounding at Multiple Granularities in Documents

B. V. Kasuba, P. Chaudhuri, G. Ramakrishnan

2025 arXiv preprint

IIT Bombay

7 HindiOCR-VLM

Adapting Vision-Language Models for Hindi OCR (HindiOCR-VLM)

S. Bhattacharyya, S. Ghosh, P. Deb, A. Mondal, C. V. Jawahar

2025 Conference paper

IIIT Hyderabad

8 Devanagari Tables

Incorporate Structure and Content for Devanagari Table Extraction

A. R. Dumada, S. G. Shah

2026 IJSAT 17(1)

Deogiri Institute, Chh. Sambhajinagar

IndicDLP

IndicDLP: A Foundational Dataset for Multi-Lingual and Multi-Domain Document Layout Parsing — Nath et al., 2025, IIT Madras & IIIT Hyderabad

PROBLEM

Layout datasets are mostly English; models do poorly on Indic pages.

KEY IDEA

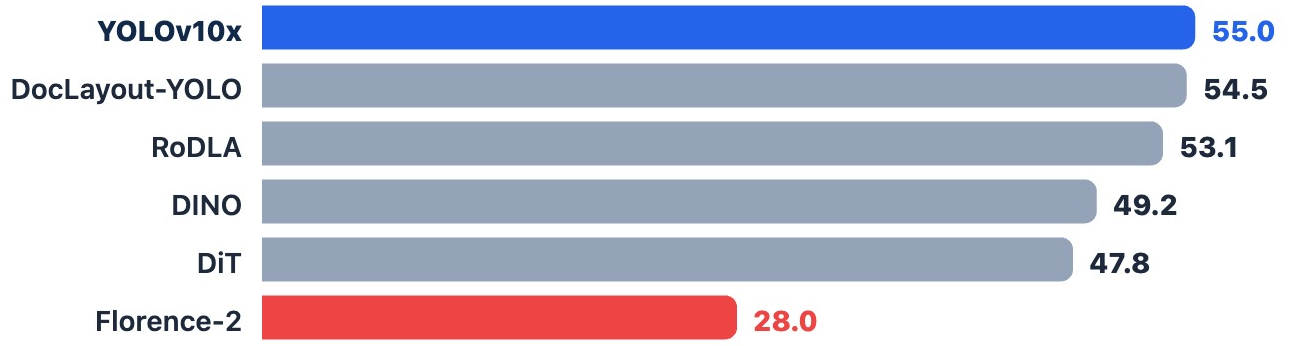
A huge human-labelled layout dataset across 12 languages and 12 document types; benchmark six detectors.

DATA & MODELS

~120K pages · 12 domains · 42 region labels

YOLOv10x · DocLayout-YOLO · RoDLA · DINO · DiT · Florence-2

RESULT mAP@[50:95] on IndicDLP



55.0 best mAP

–20 to –25 mAP on unseen scripts

METHODOLOGY WORKFLOW

1. **Collect** — 12 domains
2. **Annotate** — 50-person team
3. **Review** — Superchecks
4. **Split** — 80/10/10
5. **Train** — 6 detectors
6. **Evaluate** — mAP

Limitation: Vision-only models; rectangular boxes only.

Our project: Our dataset source (AIKosh); its YOLO results are the reference for YOLOv8s vs Faster R-CNN.

PLATTER

PLATTER: A Page-Level Handwritten Text Recognition System for Indic Scripts — Kasuba et al., 2025, IIT Bombay

PROBLEM

Indic handwriting work is word-level, compared inconsistently, with little page-level data.

KEY IDEA

Two stages: find words (DBNet), then read them with one of six recognisers.

DATA & MODELS

New CHIPS dataset: ~25K pages, 10 languages

DBNet · CRNN · MASTER · PARSeq · +3 more

RESULT **Word accuracy (%): clean words vs words cut from pages**



99.7% detection F1

73% → 38% CRNN on page crops

METHODOLOGY WORKFLOW

1. **Page** — Handwritten
2. **Detect** — DBNet words
3. **Crop** — Each word
4. **Recognise** — 1 of 6 models
5. **Output** — Text on page

Limitation: Synthetic pages; language must be named by the user.

Our project: Detection precedes recognition; our layout output can feed an OCR stage.

UniTabBank

UniTabBank: A Large Scale Multi-Lingual, Multi-Layout, Multi-Type, Multi-Format Dataset for Table Detection — Mondal et al., 2026, IIIT Hyderabad

PROBLEM

Table datasets are English, single-domain, clean PDFs; detectors fail on real documents.

KEY IDEA

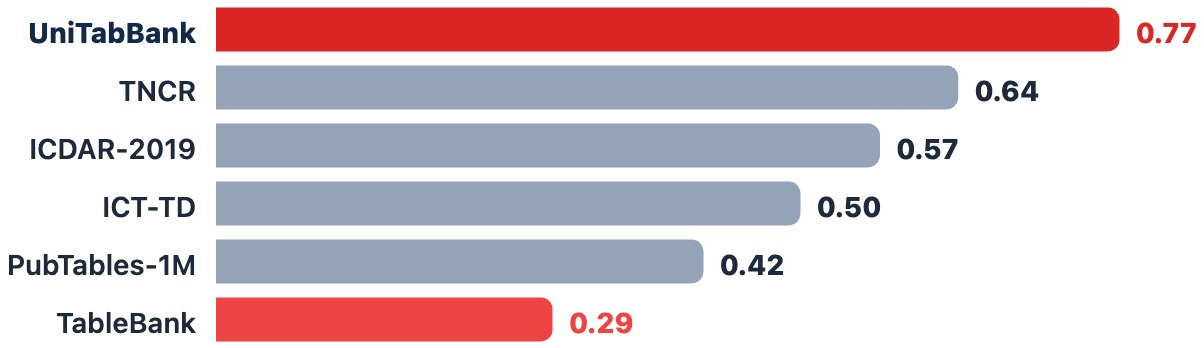
One diverse table dataset (PDFs, scans, photos) auto-labelled then fixed by humans; YOLOv11 detector.

DATA & MODELS

55K pages · 82K tables · 28 languages

UniTabDet (YOLOv11) · DocLayout-YOLO · TATR

RESULT AP on unseen UNLV data, by training dataset



8 of 9 benchmarks beaten

0.77 vs 0.42 AP on UNLV

METHODOLOGY WORKFLOW

1. **Collect** — 28 languages
2. **Auto-label** — DocLayout-YOLO
3. **Fix** — Annotators
4. **Recheck** — 20% sample
5. **Train** — UniTabDet
6. **Test** — 9 benchmarks

Limitation: Borderless tables and degraded scans.

Our project: Tables are one of our layout classes; YOLO-family detectors work well here.

Indic Table QA

Table Question Answering for Low-resourced Indic Languages — Pal et al., 2024, University of Amsterdam

PROBLEM

No table QA data for low-resource languages; translating English tables is error-prone.

KEY IDEA

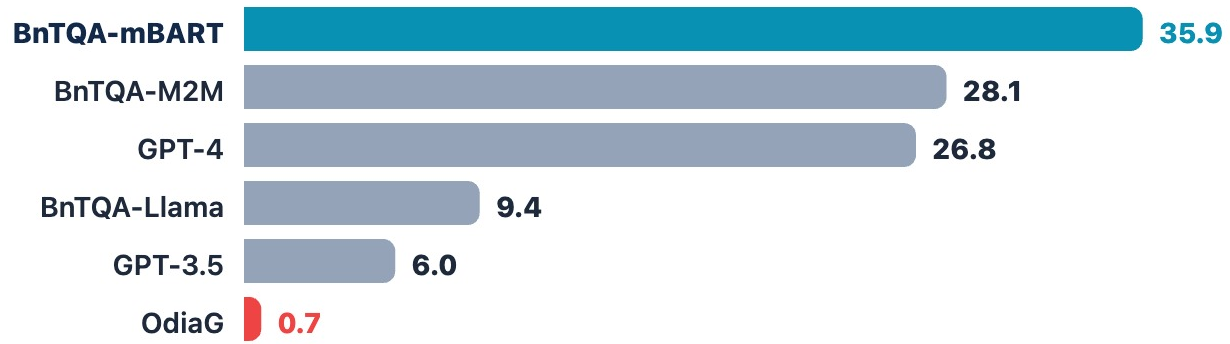
Auto-generate questions and answer tables from Bengali / Hindi Wikipedia tables using SQL.

DATA & MODELS

BanglaTabQA 2M · HindiTabQA 643K samples

mBART · M2M-100 · Llama-7B · vs GPT-4

RESULT Bengali test set — table exact match (%)



35.9% vs 26.8% mBART vs GPT-4

2M + 643K Bengali + Hindi samples

METHODOLOGY WORKFLOW

1. **Tables** — Wikipedia
2. **SQL** — Templates
3. **Question** — SQL2NQ
4. **Answer** — Run SQL
5. **Filter** — Quality check
6. **Train** — mBART...

Limitation: Needs web tables; no multi-table queries; small test sets.

Our project: Detected tables can feed table QA on Indian-language data.

LayoutXLM

LayoutXLM: Multimodal Pre-training for Multilingual Visually-rich Document Understanding — Xu et al., 2021, Microsoft Research Asia

PROBLEM

Document AI models were English-only; text-only multilingual models ignore layout.

KEY IDEA

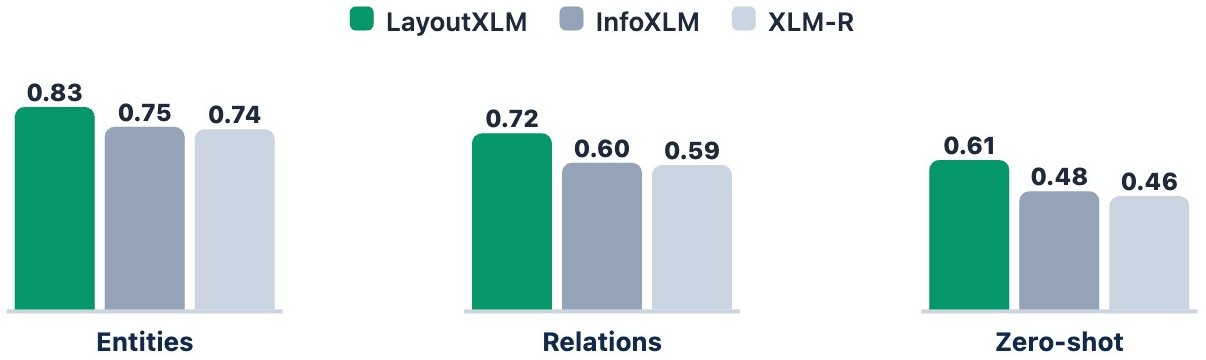
Pre-train one model on text + layout + image across 53 languages; new XFUND benchmark.

DATA & MODELS

30M docs, 53 languages · XFUND: 1,393 forms, 7 languages

LayoutLMv2 · InfoXLM · MMVLM · TIA · TIM

RESULT **Average F1 on XFUND + FUNSD (large models)**



0.83 vs 0.74 entity F1 vs XLM-R

Zero-shot transfers to new languages

METHODOLOGY WORKFLOW

1. **Page** — Document
2. **OCR** — Words + boxes
3. **Embed** — Text+layout+image
4. **Transformer** — Pre-train
5. **Fine-tune** — Entities

Limitation: No Indic language in XFUND; needs OCR; large model.

Our project: Layout regions help document understanding; Indic scripts not covered.

DRISHTIKON

DRISHTIKON: Visual Grounding at Multiple Granularities in Documents — Kasuba et al., 2025, IIT Bombay

PROBLEM

Document QA gives answers but not where on the page they come from.

KEY IDEA

LLM gives the answer; a matching algorithm finds it on the page at block, line, word or point level.

DATA & MODELS

MGVG: 70 circulars · 509 QA pairs

DocTR / Surya OCR **Llama-3.1-8B** **Region matching**

RESULT Grounding F1 (%) by level



69.1% line-level F1 (best)

5.2% Qwen2.5-VL alone

METHODOLOGY WORKFLOW

1. Input Page + question → **2. OCR** Text blocks → **3. Answer** Llama-3.1-8B → **4. Match** Fuzzy score → **5. Ground** Block→point

Limitation: Depends on OCR; single page; small test set.

Our project: Our layout regions can serve as blocks for answer grounding.

HindiOCR-VLM

Adapting Vision-Language Models for Hindi OCR (HindiOCR-VLM) — Bhattacharyya et al., 2025, IIIT Hyderabad

PROBLEM

Indian-script OCR is hard (ligatures, diacritics); each text type needs a separate model.

KEY IDEA

LoRA-tune GOT OCR 2.0 for Hindi, learning step by step: words → lines → blocks → pages.

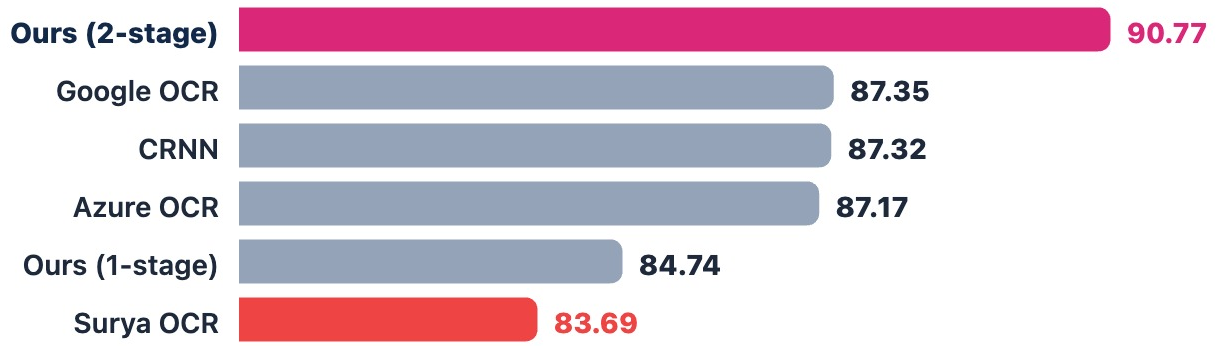
DATA & MODELS

Printed Hindi: 1.1M words → 3.6K pages (+ handwritten, scene)

GOT OCR 2.0 | LoRA (rank 8) | CRAFT / YOLO

RESULT Printed Hindi — word recognition rate (%)

Axis starts at 80



90.77 vs 87.35 vs Google OCR

56% faster 350K vs 800K steps

METHODOLOGY WORKFLOW

1. **Base** — GOT OCR 2.0
2. **LoRA** — Rank 8
3. **Words** — 50K steps
4. **+ Lines** — Fast transfer
5. **+ Pages** — From 200K
6. **Done** — 350K steps

Limitation: Weak on mixed Hindi-English lines; scene text trails Google OCR (67.15 vs 74.38 WRR); Hindi only (Marathi WRR 36.70).

Our project: Page-level OCR stage that can follow our layout detection for Devanagari text.

Devanagari Tables

Incorporate Structure and Content for Devanagari Table Extraction — Dumada & Shah, 2026, Deogiri Institute, Chh. Sambhajinagar

PROBLEM

Table extractors perform poorly on Devanagari: ligatures, noisy scans, scarce data.

KEY IDEA

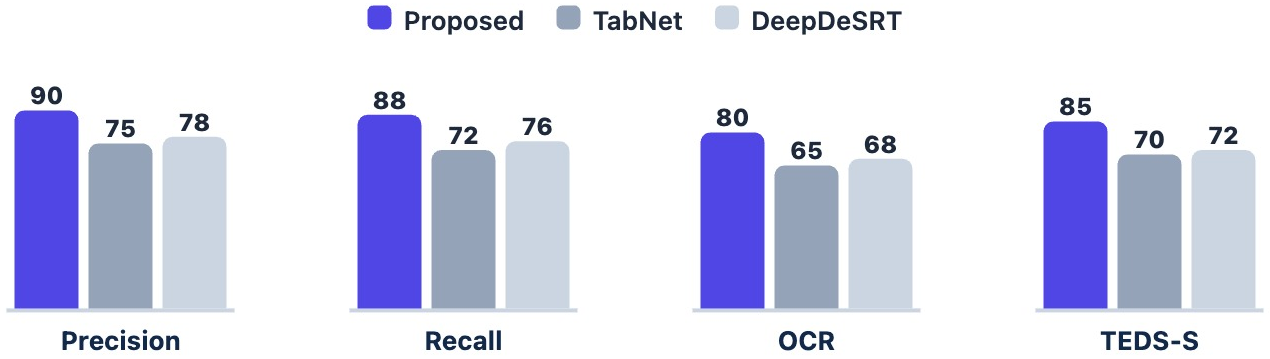
Combine table structure (lines, cells) with OCR content checks using weighted fusion.

DATA & MODELS

500+ annotated Devanagari documents

Hough deskew | Contours | Tesseract | Weighted fusion

RESULT Proposed vs baselines (%) — self-reported



85% TEDS-S (vs 70% TabNet)

90% structural precision

METHODOLOGY WORKFLOW

1. **Input** — Document
2. **Pre-process** — Denoise, deskew
3. **Structure** — Lines & cells
4. **OCR** — Tesseract
5. **Fusion** — Weighted
6. **Output** — Table

Limitation: Dataset, baseline adaptation and fusion weights not described; no ablation; results are self-reported.

Our project: Supporting reference for the table-structure stage; its claims need independent validation.

LITERATURE SURVEY TABLE

Summary: Strengths of Papers

S.N.	Paper	What is good	Innovations	Assumptions
1	**IndicDLP** Nath et al., 2025 · arXiv preprint	Largest human-labelled Indic layout set	42 fine labels (jumpline, flag…) + UED-mini pre-training	Rectangular boxes are enough
2	**PLATTER** Kasuba et al., 2025 · arXiv preprint	Fair comparison of 6 HTR models, 10 languages	Language-agnostic detector + CHIPS dataset	Synthetic pages behave like real ones
3	**UniTabBank** Mondal et al., 2026 · IEEE/CVF WACV	28 languages; scans + photos; generalises	Semi-automatic labelling	Script barely affects table detection
4	**Indic Table QA** Pal et al., 2024 · arXiv preprint	Cheap, automatic data; small models beat GPT-4	SQL → question model + quality filter	Wikipedia tables represent real ones
5	**LayoutXLM** Xu et al., 2021 · arXiv preprint	Text + layout + image together	Language-free token boxes; XFUND	Good OCR is available
6	**DRISHTIKON** Kasuba et al., 2025 · arXiv preprint	Shows where the answer is → trust	Multi-level grounding + MGVG	Answer appears word-for-word in OCR
7	**HindiOCR-VLM** Bhattacharyya et al., 2025 · Conference paper	One model for printed, handwritten, scene	Progressive learning, 56% faster	English/Chinese encoder transfers
8	**Devanagari Tables** Dumada & Shah, 2026 · IJSAT 17(1)	Targets Devanagari tables directly	Structure + content fusion	Tesseract is accurate enough

LITERATURE SURVEY TABLE

Summary: Limitations of Papers

Paper	Weaknesses	Missing aspects	Assumptions
1. IndicDLP Nath et al., 2025 · arXiv preprint	Vision-only; –20–25 mAP on unseen scripts	Polygon boxes; text cues	Look alone tells region type
2. PLATTER Kasuba et al., 2025 · arXiv preprint	Big accuracy drop on page crops	Auto script detection; real pages	User names the language
3. UniTabBank Mondal et al., 2026 · IEEE/CVF WACV	Borderless tables, degraded scans	Row / cell structure	Auto labels nearly correct
4. Indic Table QA Pal et al., 2024 · arXiv preprint	Small test sets (165 Bengali)	Multi-table, complex queries	Clean tables, no OCR noise
5. LayoutXLM Xu et al., 2021 · arXiv preprint	Large model; needs OCR	No Indic language	Synthetic forms look real
6. DRISHTIKON Kasuba et al., 2025 · arXiv preprint	OCR-dependent; small test set	Multi-page, mixed script	Fixed limits suit all documents
7. HindiOCR-VLM Bhattacharyya et al., 2025 · Conference paper	Mixed-language lines; scene text	Other Indic languages	In-house data is representative
8. Devanagari Tables Dumada & Shah, 2026 · IJSAT 17(1)	Self-reported; no ablation	Method details, public data	Baselines adapted fairly

WHAT IS MISSING

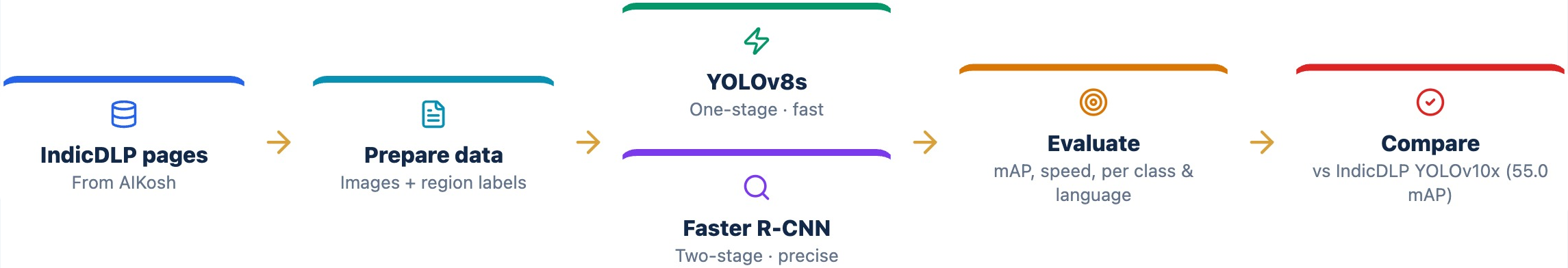
Research Gaps

IMPROVEMENTS SUGGESTED BY US		POSSIBLE EXTENSIONS BY US
⚠ Only big models benchmarked on IndicDLP	→ ✓ **Compare lightweight YOLOv8s vs Faster R-CNN**	**Auto script detection** Pick the right OCR per region
⚠ –20–25 mAP on unseen scripts	→ ✓ **Train on all languages; report per-language mAP**	**Multi-page documents** Grounding & QA beyond one page
⚠ Borderless tables, degraded scans fail	→ ✓ **Add blur / noise augmentation**	**Text + layout for Indic** LayoutXLM-style model for Indian scripts
⚠ Layout, OCR, tables, QA built separately	→ ✓ **Output boxes the next stages can use**	**Lightweight deployment** Small models on mobile / edge

OUR PROJECT

Our Proposed Idea

NEW APPROACH — WORKFLOW



OUR IMPROVEMENT IDEA

Test a lightweight YOLOv8s against Faster R-CNN on IndicDLP — a pair the IndicDLP paper did not test — with output ready for the next stages.

EXPECTED OUTCOME

- Speed vs accuracy trade-off for Indic pages
- Which regions / scripts are hardest
- Region boxes ready for OCR & tables

NEXT STAGES

- **OCR** → HindiOCR-VLM / PLATTER
- **Tables** → UniTabBank + Table QA
- **Grounding** → DRISHTIKON

KEY LEARNINGS

Conclusion

Data is the bottleneck

Big human-labelled Indic datasets drive progress.

Detect first, then read

Two-stage pipelines win.

YOLO-family is strong

Accurate and fast for layout and tables.

Script matters

–20–25 mAP on unseen scripts.

Small tuned models win

Fine-tuned mBART beat GPT-4.

VLMs can't locate well

Qwen2.5-VL: 5.2% grounding F1.

Takeaway: layout detection is the first building block of Indian document AI — that is what our project builds.

SOURCES

References

PAPER CITATIONS

[1] O. Nath et al., "IndicDLP: A Foundational Dataset for Multi-Lingual and Multi-Domain Document Layout Parsing," arXiv:2512.20236, 2025.

[2] B. V. Kasuba et al., "PLATTER: A Page-Level Handwritten Text Recognition System for Indic Scripts," arXiv:2502.06172, 2025.

[3] A. Mondal et al., "UniTabBank: A Large Scale Multi-Lingual, Multi-Layout, Multi-Type, Multi-Format Dataset for Table Detection," IEEE/CVF WACV, pp. 6486–6495.

[4] V. Pal et al., "Table Question Answering for Low-resourced Indic Languages," arXiv:2410.03576, 2024.

[5] Y. Xu et al., "LayoutXLM: Multimodal Pre-training for Multilingual Visually-rich Document Understanding," arXiv:2104.08836, 2021.

[6] B. V. Kasuba et al., "DRISHTIKON: Visual Grounding at Multiple Granularities in Documents," arXiv:2506.21316, 2025.

[7] S. Bhattacharyya et al., "Adapting Vision-Language Models for Hindi OCR," IIIT Hyderabad, hindiocr-vlm.github.io.

[8] A. R. Dumada, S. G. Shah, "Incorporate Structure and Content for Devanagari Table Extraction," IJSAT, vol. 17, no. 1, 2026.

OTHER SOURCES

- IndicDLP dataset — AIKosh
- indicdlp.github.io
- github.com/iitb-research-code/doctr
- aka.ms/layoutxlm
- github.com/kolk/Low-Resource-TableQA-Indic-languages
- hindiocr-vlm.github.io

Thank You

Questions and discussion

Prashant Jha | Aditya Soni | Riddhi Patil | Mariam Badure

IndicDoc AI | Literature Survey